Experiment Update: Agentic Processing of ARPES Data with Claude Sonnet 5.5 (medium effort) and a Local ARPES Viewer

Session date: 30 October 2026. Model: `claude-sonnet-5-5`, effort level "medium" (as specified by the user). Sample identity and lattice constants are withheld; the lattice is referred to by its parameters a, b, c only. Light polarisations (LH, LV, CR, CL) are retained.

1. Summary

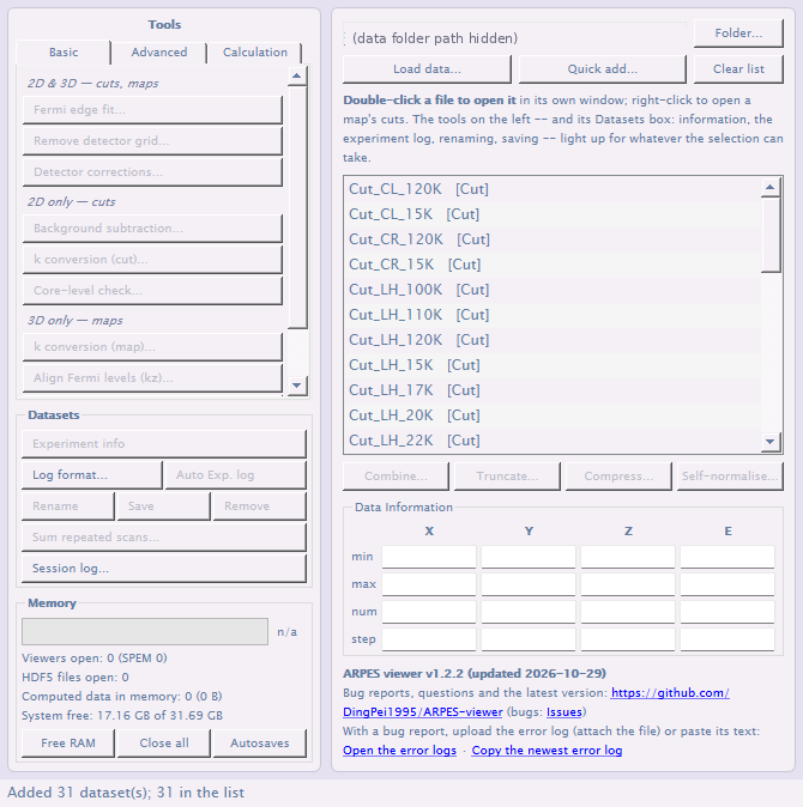
- A local, user-controlled agentic workflow processed one beamtime folder (22 slit cuts + 4 Fermi-surface maps, hv = 79 eV) with the lab's own PyQt5 ARPES viewer. **No raw data and no code left the user's computer**; the only visual material shared with the model were screenshots and finished figures.
- Completed: per-cut intensity normalisation, Fermi-edge fits and E-E_F calibration, momentum conversion (maps and cuts), Brillouin-zone overlay, iso-energy figures (0 to -0.7 eV), LH/LV/CR/CL cut comparison with linear and circular dichroism, a 17-panel LH temperature series, and a quantitative MDC two-peak separation vs temperature.
- Main physical observation: the separation of the two MDC peaks at E-E_F = -0.1 eV (Delta k) decreases from 0.211 1/A (15 K) to 0.172 1/A (120 K); the point measured after cooling back to 20 K (0.180 1/A) does **not** return to the warming value at 20 K (0.208 1/A). A geometry cross-check (Section 7) shows this is unlikely to be a sample-holder rotation artefact, but it is not yet settled.
- Cost: about **7.3 USD** of API-equivalent usage for the whole data-processing session (about 43 M cache-read tokens, 0.4 M cache-write tokens, 0.14 M output tokens); details in Section 8. Plan-percentage usage must be read by the user from the account usage page.

2. Ground rules agreed at the start

1. Data processing is done locally. Raw data and code are never uploaded to any server; visual judgement is done through screenshots only.
2. The agent works on a second monitor and the user only talks to it in the chat window (the user does not click or type for it).
3. The viewer program's own functions are the default for every operation; anything custom (own NumPy code) is declared explicitly.
4. When a parameter is uncertain (e.g. the Gamma point), the data are opened and the user is asked.
5. Real runs keep only the original data and the final-step result in the viewer's main list (intermediates and their autosave files are removed).
6. Figures default to "Smooth" ticked with blur 0.
7. Reports are in English; sample identity is hidden.

3. Dataset

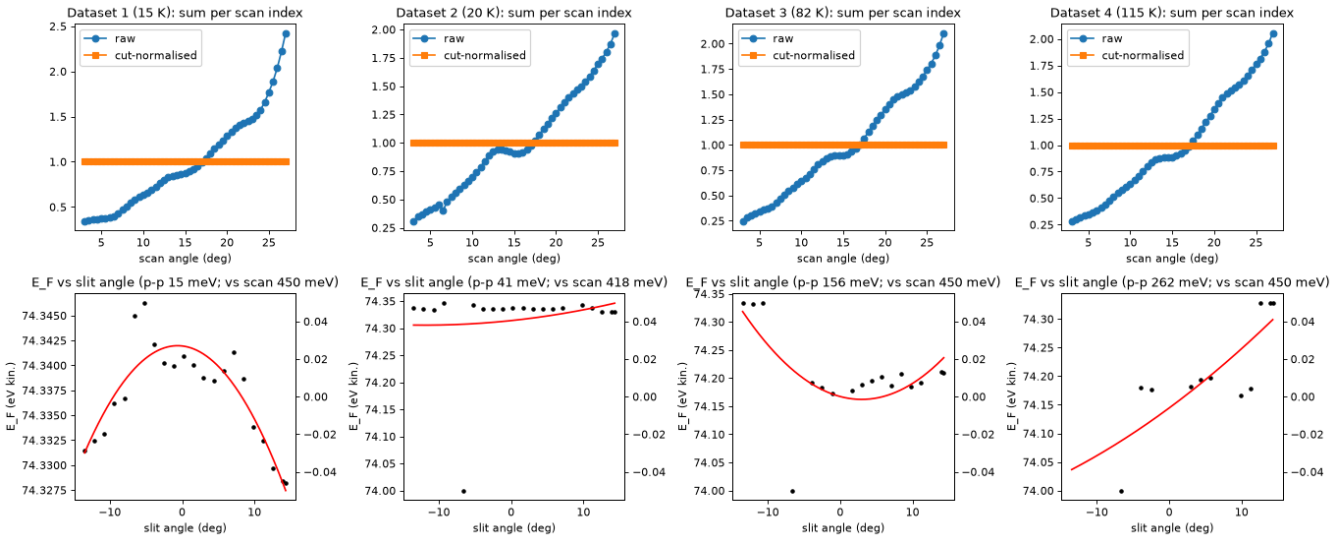
Item	Content
Slit cuts	22: LH at 16 temperatures (15, 17, 20, 22, 26, 30, 35, 40, 50, 60, 70, 80, 90, 100, 110, 120 K); LV, CR, CL at 15 K and 120 K
Fermi-surface maps	4 “FineMap” scans at 15, 20 (taken after the warming loop, i.e. cooled back), 82 and 115 K; 49 scan angles x 609 slit angles x 251-271 energy points
Photon energy	79 eV, work function 4.59 eV (file header)
Excluded	5 additional map folders (“Map*“) removed from the list at the user’s request



All 31 datasets listed in the viewer’s main panel (path field hidden).

4. Processing steps for the four maps

4.1 Intensity asymmetry along the scan axis and normalisation. The summed intensity per scan index rises monotonically along the scan axis (last/first ratio 6.4-8.5 for the four maps), so each constant-scan-angle plane (slit x energy) was normalised to equal total area with the viewer’s *Self-normalise* dialog (suffix `_s_nor`). An initial scripted version with an extra rescale was discarded because the user wanted the program function.

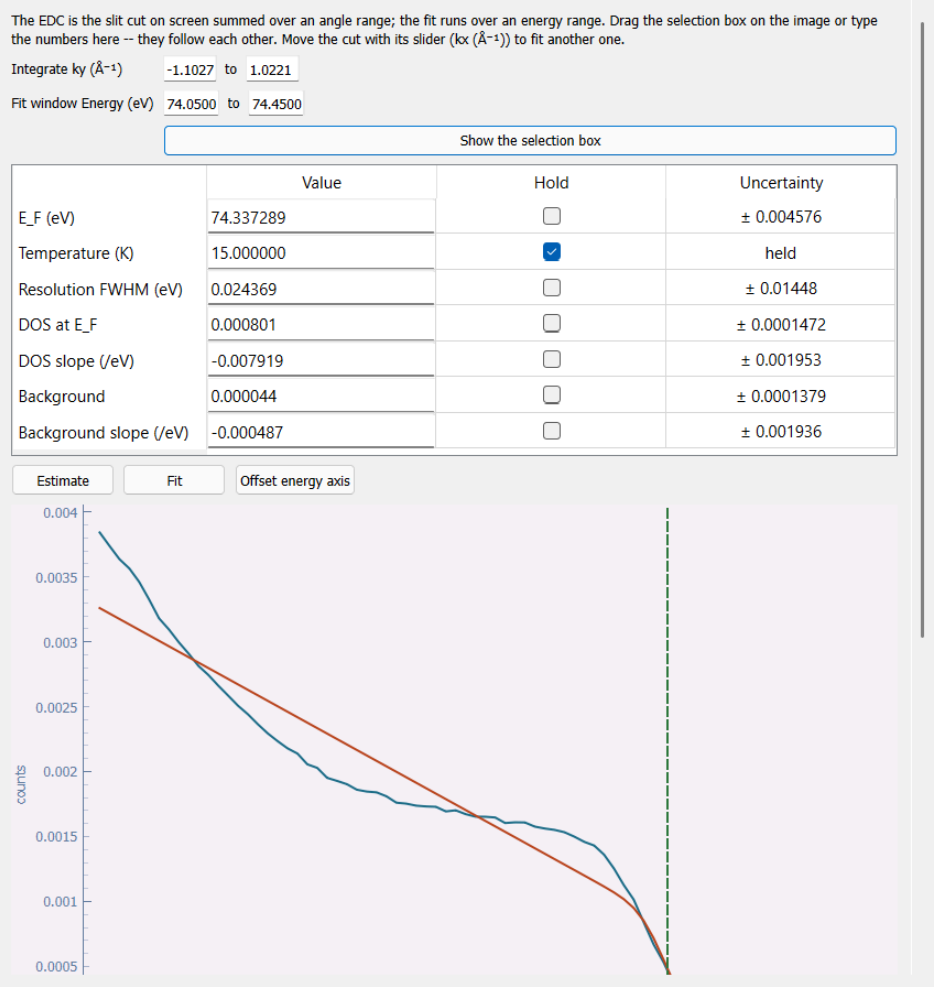


Left: intensity per scan index before/after cut normalisation. The lower row shows early exploratory E_F-vs-slit-angle diagnostics (superseded by the program's own Fermi fit).

4.2 Fermi-surface flatness. The slit-angle dependence of E_F of the coldest map is about 15 meV peak-to-peak, comparable to the fitted energy resolution (14-29 meV). By the rule set by the user (variation of about the resolution means no flattening), the raw geometry was kept.

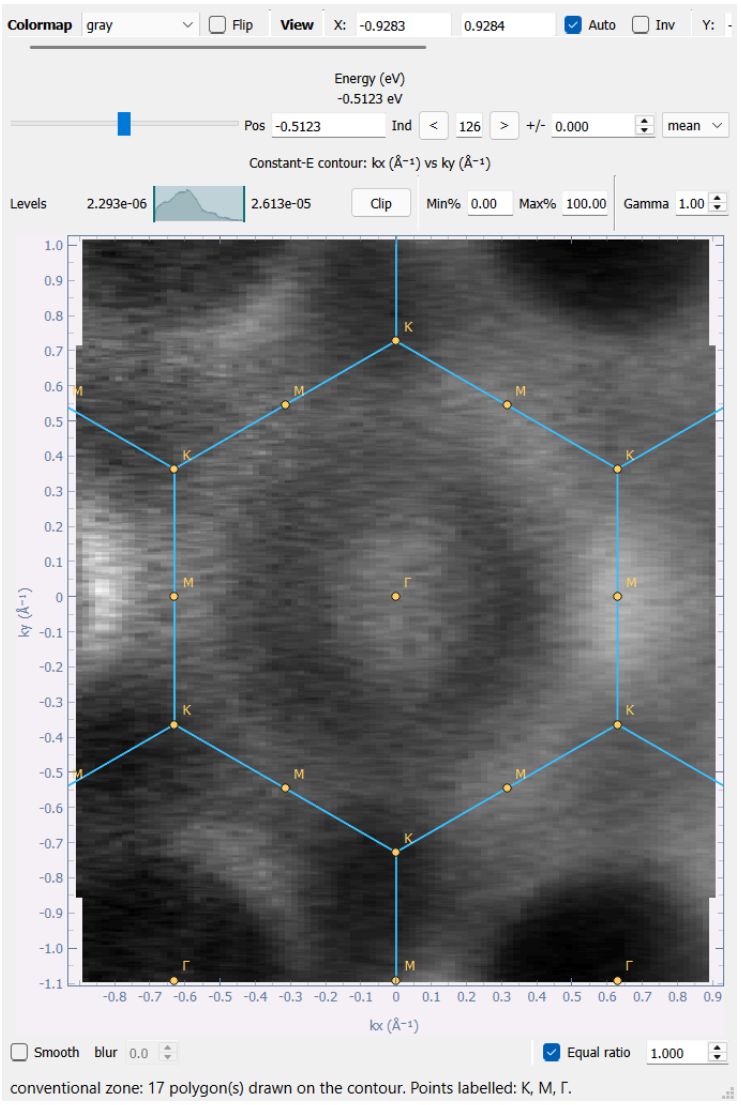
4.3 Gamma point and momentum conversion. The origin was set by the user to scan angle 15.0 deg and slit angle 0.96 deg for all four maps (same Gamma for all). Converted with the program's *k conversion (map)* dialog; because the scan axis has only 49 points, the output kx grid was set to 100 points (user rule), ky and E grids as measured. Result: kx in [-0.919, 0.919] 1/A, ky in [-1.103, 1.022] 1/A.

4.4 Fermi edge and energy offset. The program's *Fermi edge fit* was run on each converted map (energy window 74.05-74.45 eV kinetic, temperature held at the cryostat value) and the energy axis offset (whole map) to E-E_F = 0. Fitted E_F is about 74.33-74.34 eV (kinetic) with 2-5 meV uncertainty; fitted resolution 14-29 meV.

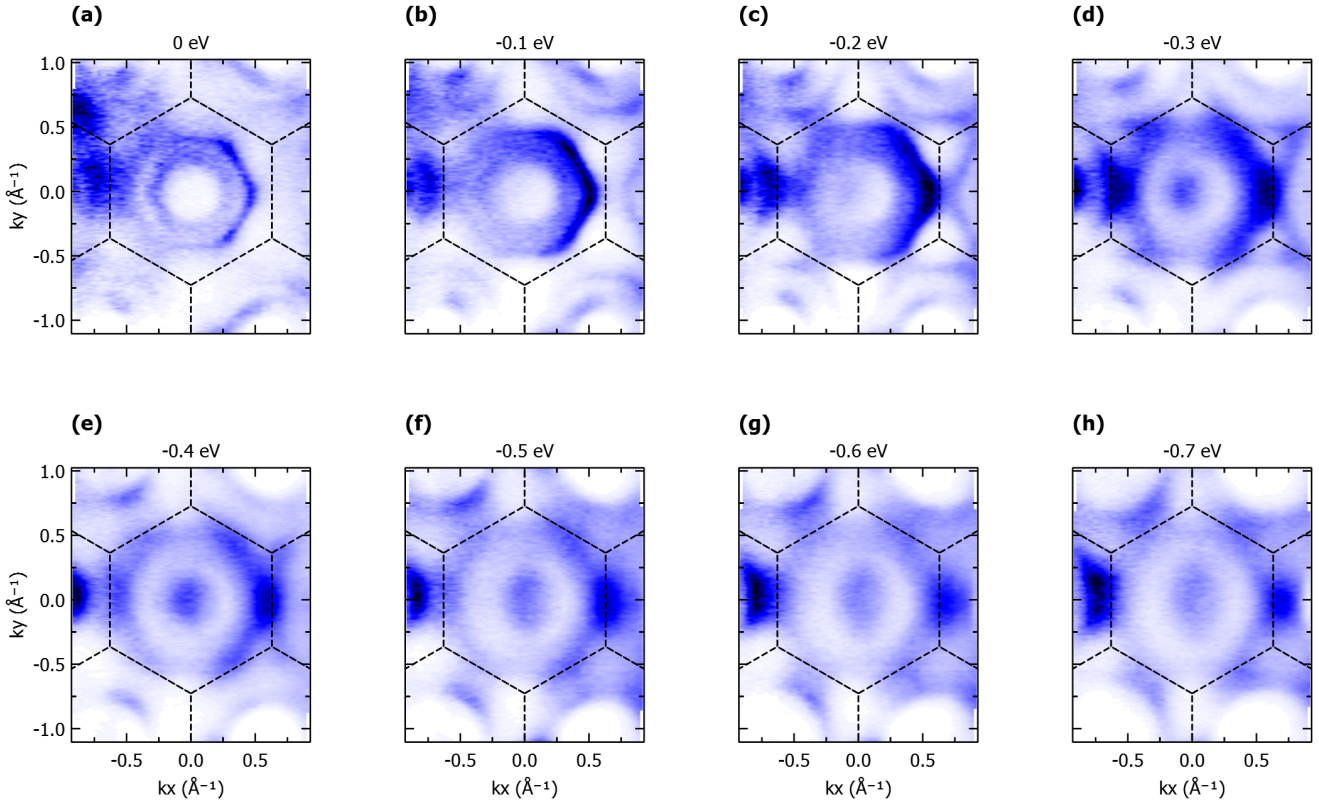


Program’s Fermi-edge fit panel for the 15 K map.

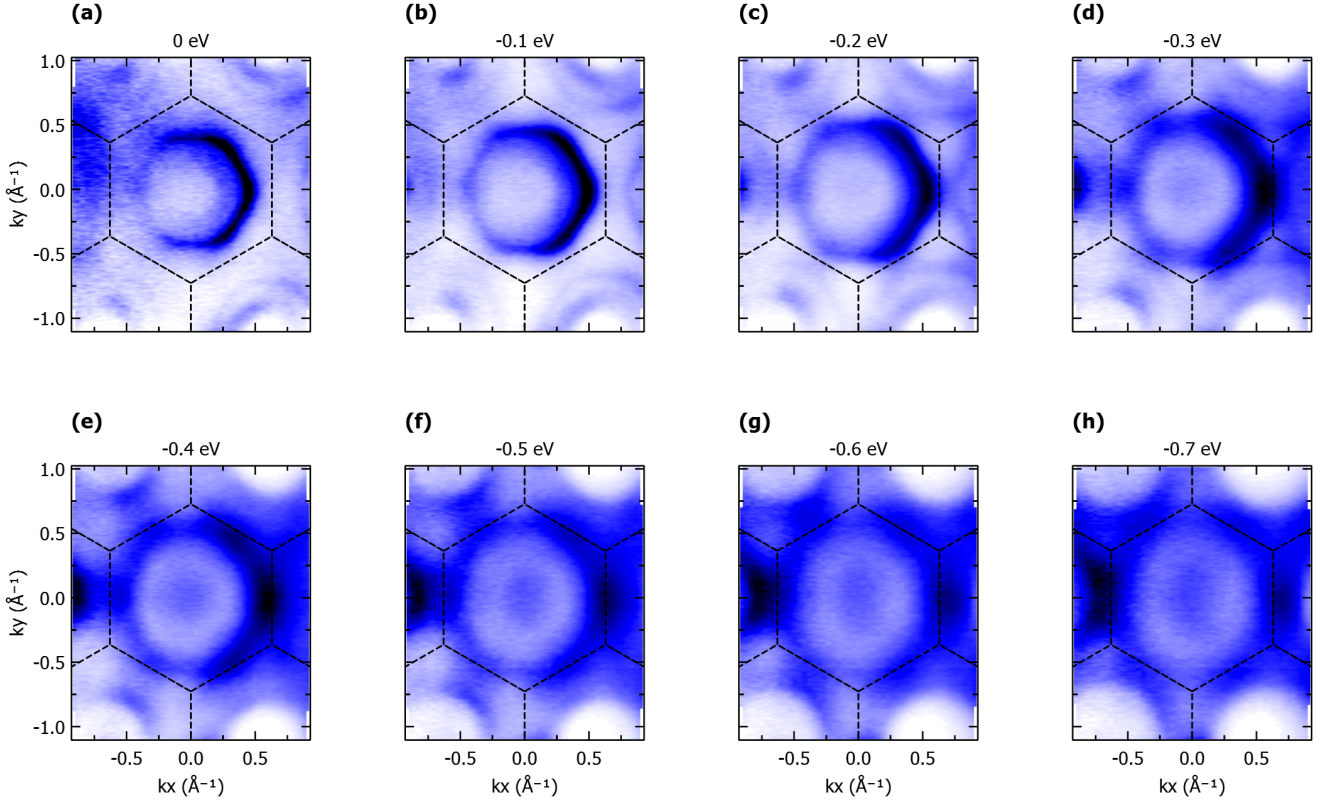
4.5 Brillouin zone and iso-energy figures. The 3D-crystal Brillouin zone was overlaid with the program’s *Brillouin zone* dialog (cut plane (001), in-plane rotation 30 deg, hv = 79 eV taken as the Gamma plane; lattice parameters a, b, c supplied by the user and not reproduced here). Iso-energy figures (0 to -0.7 eV in 0.1 eV steps, +-20 meV integration, 4 x 2 layout, blue colormap flipped, 300 dpi) were generated through the program’s slice-series and figure composer; level and gamma settings were tuned panel-wise at the user’s request (e.g. Min 0, Max 100, gamma 2 for the 15 K figure, Smooth on, blur 0).



Brillouin zone drawn on a constant-energy contour (program screenshot, grey colormap).



Iso-energy figure of the 15 K map.

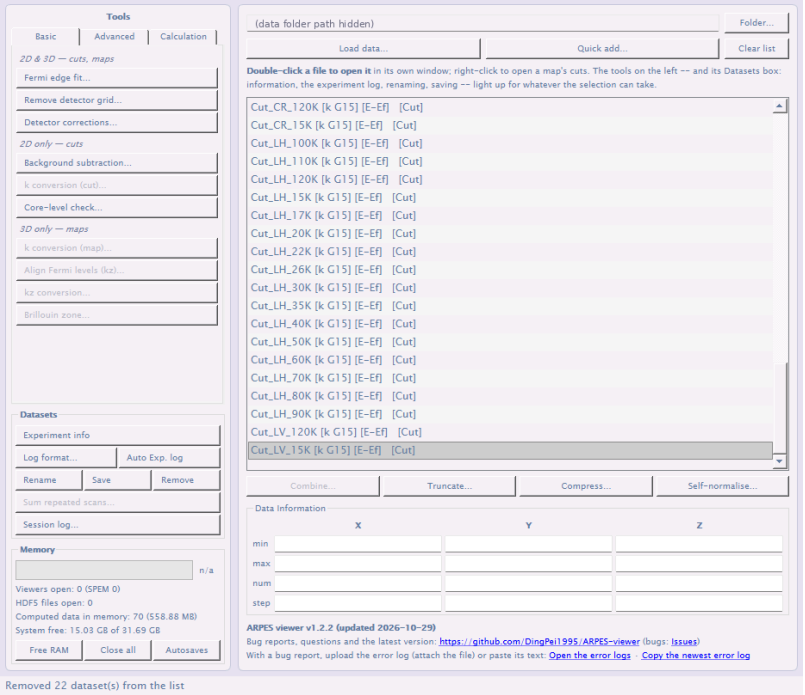


Iso-energy figure of the 115 K map.

5. Processing steps for the 22 cuts

1. *k conversion (cut)*: first attempt with deflector angle 0 and Gamma slit angle 0.8 deg; then, with the true geometry, cut deflector angle 23.5 deg and Gamma deflector angle 15 deg (the user's decision: keep this version, remove the other).

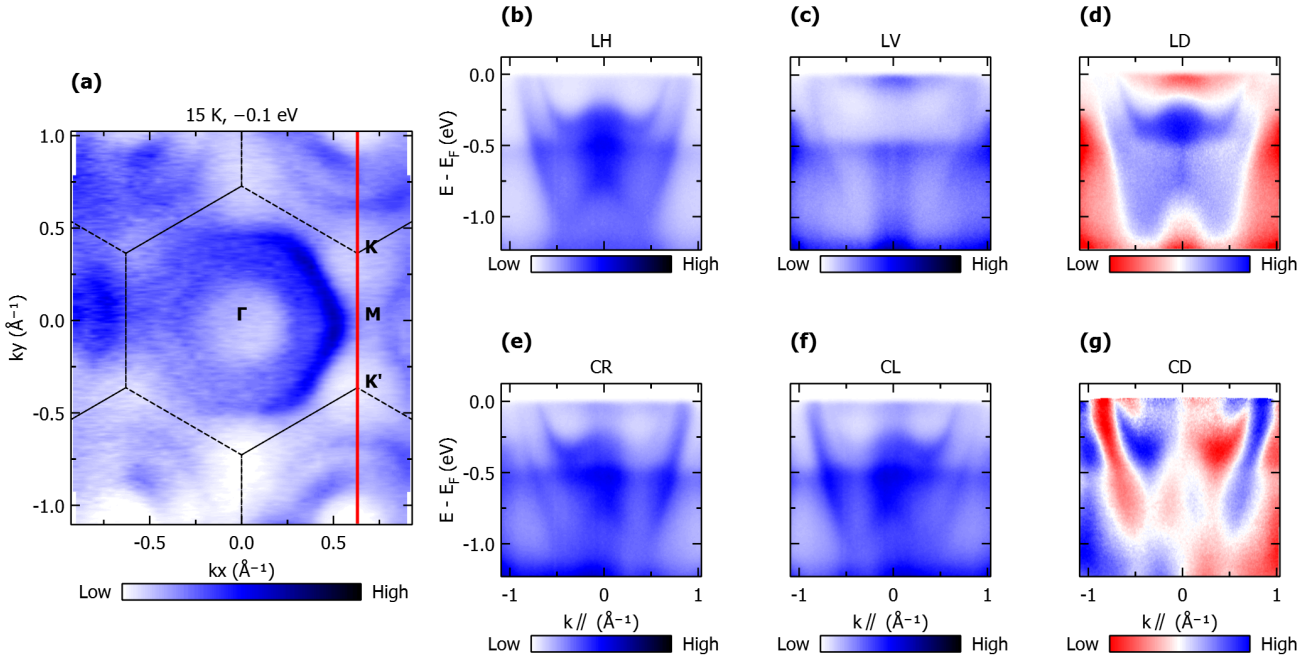
2. *Fermi edge fit* (window 74.05-74.45 eV, T held at the file-name temperature) and *Offset energy axis*: E_F (kinetic) 74.326-74.356 eV, fit errors 1-16 meV (largest for the 100-120 K LH cuts).
3. Total-intensity normalisation per cut (program *Self-normalise*) for all cuts used in comparisons.
4. Linear dichroism LD = LH - LV (LV scaled to the same total) and circular dichroism CD = (CR - CL)/(CR + CL) with the program's cut-arithmetic dialog, symmetric red-white-blue scale.



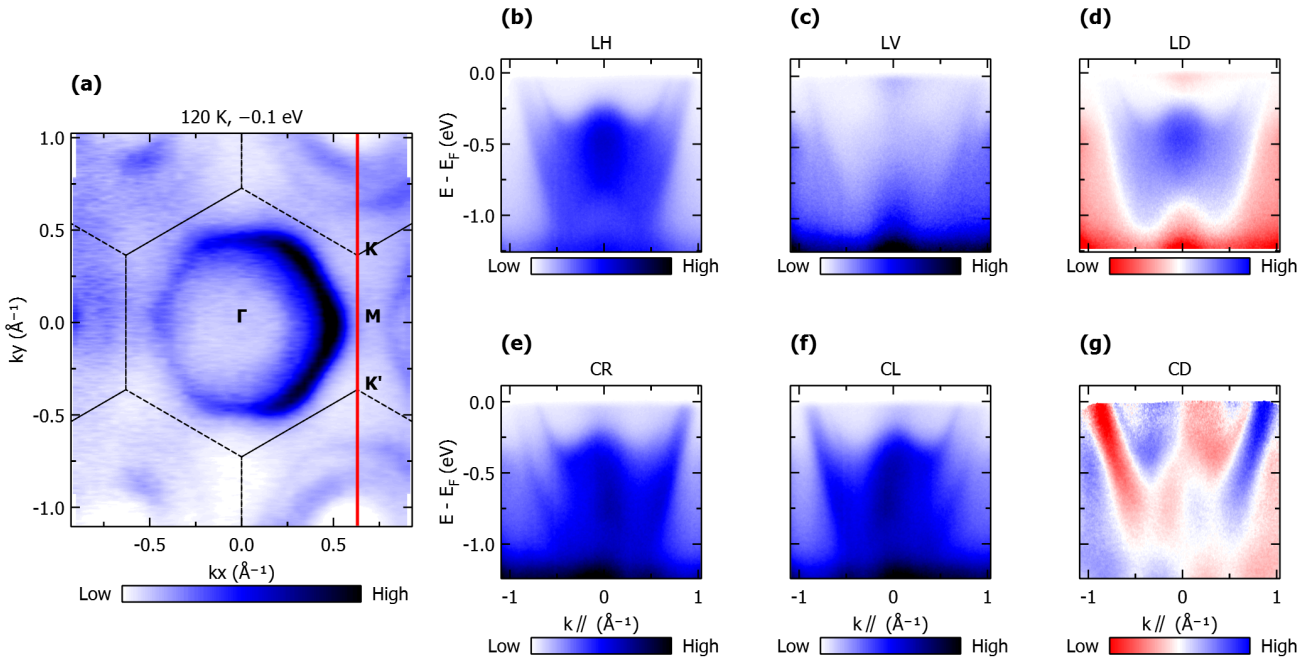
Main list after cut conversion: original cuts plus the final `[k G15] [E-Ef]` cuts (path field hidden).

6. Results

6.1 Cut comparison at 15 K and 120 K. Panel (a): iso-energy contour at -0.1 eV of the corresponding FineMap with the Brillouin zone and a red guide line through K-M-K' marking the cut position (cut position 0.644 +- 0.013 1/A from correlation with the map; guide line drawn on the zone boundary at 0.63 1/A). Panels (b)-(g): LH, LV, LD (top row), CR, CL, CD (bottom row). Raw cuts share a colour scale per pair; dichroism panels use a symmetric red-white-blue scale.

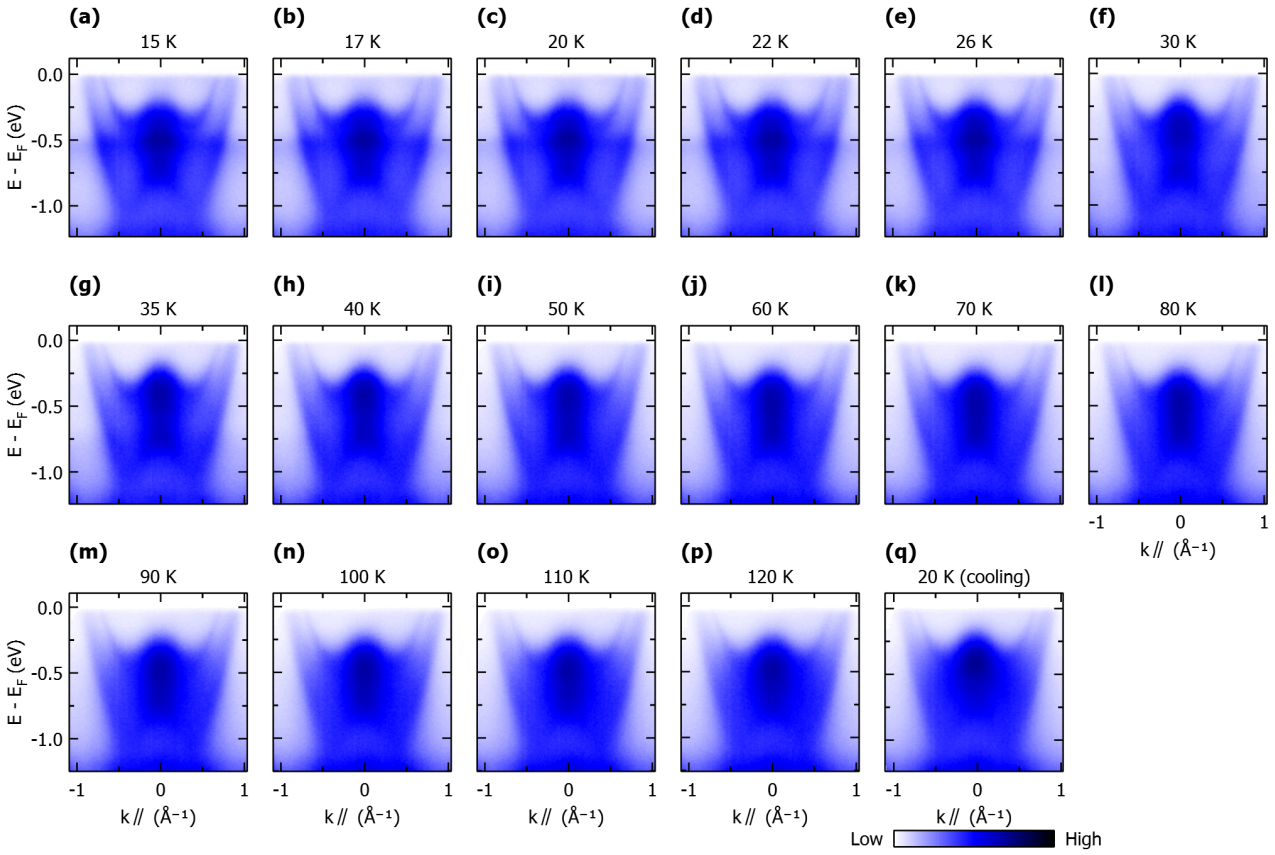


15 K: map contour, LH/LV/LD and CR/CL/CD cuts.



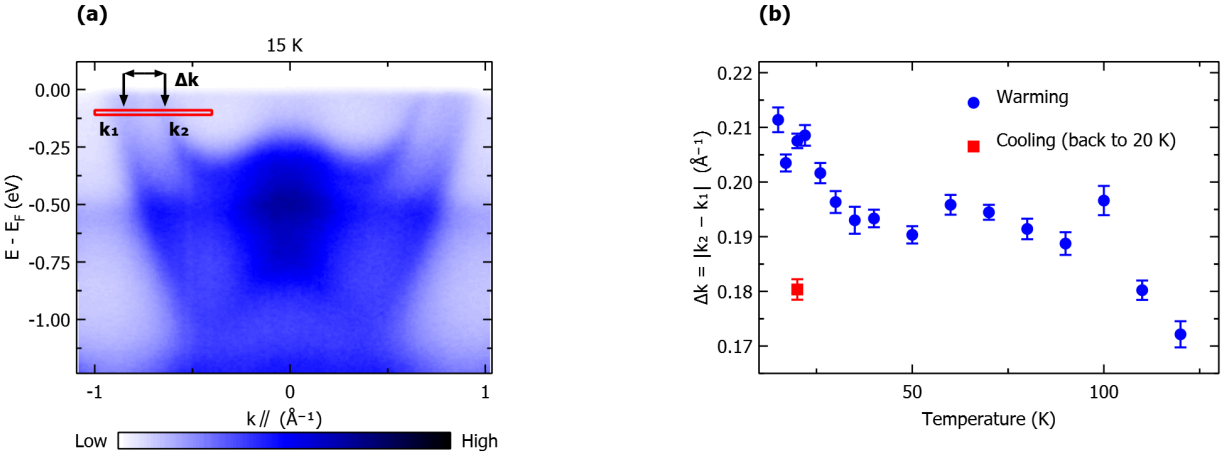
120 K: same layout (map: 115 K).

6.2 LH temperature series. Sixteen warming cuts in ascending temperature, followed by (q) the cut extracted at the same position from the 20 K map taken after the loop (cooling). Common colour scale; the cooling cut is scaled by a factor 1.012 so that its mean intensity over a common window equals that of the 20 K warming cut (own NumPy step).



LH cuts versus temperature; last panel: cooling.

6.3 MDC peak separation vs temperature. MDCs were taken at E-E_F = -0.1 eV integrating +-10 meV, k from -1.0 to -0.4 1/A, and fitted with two Lorentzians plus a linear background (unweighted, the program's fit engine `peaks.fit_line`; each fit seeded from the previous temperature). Delta k = |k2 - k1|; its error is the quadrature sum of the two fit-covariance errors (statistical only).



k₁, k₂: positions of the two MDC peaks at E − E_F = −0.1 eV (±10 meV); Δk = |k₂ − k₁| is their separation.

(a) 15 K cut with the fit window and peak positions k1, k2. (b) Delta k vs temperature; red square: cooling.

Branch	T (K)	k1 (1/A)	k2 (1/A)	Delta k (1/A)	Delta k error
warming	15	-0.8515	-0.6401	0.2114	0.0023
warming	17	-0.8465	-0.6430	0.2035	0.0016
warming	20	-0.8496	-0.6421	0.2075	0.0013

Branch	T (K)	k1 (1/A)	k2 (1/A)	Delta k (1/A)	Delta k error
warming	22	-0.8485	-0.6400	0.2085	0.0019
warming	26	-0.8507	-0.6491	0.2016	0.0019
warming	30	-0.8365	-0.6402	0.1963	0.0020
warming	35	-0.8441	-0.6511	0.1930	0.0025
warming	40	-0.8462	-0.6529	0.1933	0.0016
warming	50	-0.8534	-0.6630	0.1903	0.0016
warming	60	-0.8545	-0.6587	0.1958	0.0018
warming	70	-0.8525	-0.6581	0.1945	0.0014
warming	80	-0.8528	-0.6614	0.1914	0.0019
warming	90	-0.8552	-0.6665	0.1887	0.0021
warming	100	-0.8595	-0.6629	0.1966	0.0027
warming	110	-0.8485	-0.6683	0.1802	0.0018
warming	120	-0.8507	-0.6785	0.1721	0.0024
cooling	20	-0.8439	-0.6636	0.1803	0.0019

Notes: peak widths are large (FWHM about 0.12-0.28 1/A) and the background slope is steep, so peak positions and widths are partly correlated; scatter between neighbouring temperatures (about 0.005-0.01 1/A) exceeds the statistical error bars, which therefore understate the true uncertainty.

7. Cross-check: is the cooling point biased by the cut extracted from the map?

A same-position cut was extracted from the 20 K map with the program's *Arbitrary cut*. To test whether the offset between the cooling and warming 20 K points comes from a different cut geometry (for example a sample-holder rotation hysteresis), the image of each map slice was correlated with the corresponding measured cut (|k| <= 0.9 1/A, -1.1 to 0 eV) as a function of the map slice position kx:

Pair	Best kx (1/A)	Correlation	Equivalent deflector angle	vs. 23.5 deg
15 K map vs 15 K cut	0.661	0.991	23.61 deg	+0.11 deg
82 K map vs 80 K cut	0.666	0.992	23.67 deg	+0.17 deg
115 K map vs 120 K cut	0.672	0.992	23.76 deg	+0.26 deg
20 K (cooled) map vs 20 K (warming) cut	0.668	0.973	23.70 deg	+0.20 deg

- All pairs, including the controls, prefer a position about 0.02 1/A above the nominal 0.644 1/A, so the offset is systematic for map-to-cut matching and not specific to the cooling run. No extra lateral (ky) shift is found for the cooling map (best -0.01 vs 0 to -0.02 for the controls).

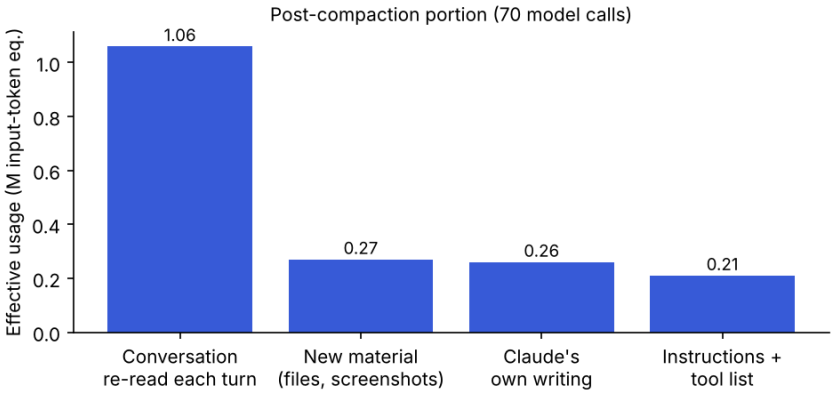
- Delta k changes by only about 0.003 per 0.01 1/A of slice position, whereas the cooling-warming gap is 0.027-0.037. Scanning the slice position over 0.60-0.72 1/A never reproduces the warming value.
- The control pairs reproduce their cut values within -0.011 to +0.014 (method scatter). The cooling gap is roughly 2-3 times larger, and its image correlation (0.973) is lower than for the controls (0.990-0.994).
- The loaded cut files carry no sample-angle metadata (only the maps do: theta/phi/tilt of the 15 K and 20 K maps differ by about 0.01 deg), so a rotation hysteresis during the cuts cannot be excluded directly; to be checked against the instrument log.

8. Token consumption and cost

Figures come from the session's own cost record (up to the context compaction) and from the session transcript (after it). They cover the whole data-processing phase and exclude the generation of this report.

Portion	Model calls	Uncached input	Cache read	Cache write	Output	Cost (USD)
From session start to compaction (recorded)	n/a	32,822	30,592,362	260,035 (1 h)	87,155 (+35,540 thinking, reported separately)	5.04 (recorded)
After compaction to start of this report (transcript)	70	172	12,728,016	134,777	51,012	about 2.2 (estimated)
Total		about 33 k	about 43.3 M	about 0.39 M	about 138 k	**about 7.3**

- A small amount of Claude Haiku 5.5 was also used (1,895 input / 24 output tokens, 0.0002 USD).
- Wall-clock time of the session at the compaction point: about 90 min; model time about 13 min and tool time about 18 min.
- **How the cost was obtained.** The recorded cost (5.04 USD) is taken from the session's cost state. The post-compaction cost was estimated by applying the same implied rate (about 1.24 USD per million "input-token equivalents", where cache read = 0.1, cache write = 2, output = 5). Public third-party list prices found online (2 USD / 10 USD per million input/output) did **not** reproduce the recorded cost, so the recorded value was preferred; please confirm the current official rates on Anthropic's pricing page.
- **Plan usage.** Plan percentage consumption is not visible to the model. Suggested entry for the final version: usage at start ____ %, at end ____ % (account usage page).
- **Where the tokens went.** About 99 % of the input is cache reads of a very long conversation (each call re-read 0.2-0.4 M tokens of context, largely screenshots and figures); fresh output is only about 0.3 % of the tokens but about 12 % of the effective cost.



Effective usage by group (post-compaction portion; instructions share estimated).

Practical savings for future runs: start a new session for each processing stage; keep screenshots small and few; read the viewer source locally with targeted searches instead of fetching it through a connector; reuse written scripts instead of re-explaining the steps.

9. Issues found and corrected during the session (for transparency)

- A 240 MB re-packaged copy of the data was created to list the files; the user objected (duplicate, unnecessary). It was deleted with user-granted permission and replaced by a live viewer instance that lists the original files directly.
- The agent removed the 22 raw cuts from the list when asked to keep only processed data; this was a misreading and they were restored.
- Cut conversion: the Gamma deflector angle is only used when the “Advanced: Gamma off this cut” option is ticked. The first deflector-23.5 attempt ignored it (recorded as 23.5 instead of 15); this was found by reading the conversion metadata, the results were discarded and redone.
- A first Fermi-edge diagnostic used a wrongly chosen window; redone with 74.0-74.45 eV and finally with the program’s own fit dialog.
- Normalisation was first scripted by hand; redone with the program’s dialog.

10. Open items

1. Verify the sample-holder angle (theta/phi/tilt) of the cuts against the instrument log to settle the hysteresis question (Section 7).
2. Decide which intermediate rows to keep in the viewer list (16 normalised LH cuts, one `Cut_LH_20K_fromMap` row and the LD/CD rows are currently kept).
3. Add statistical + systematic error estimates for Delta k (e.g. window and background variations).
4. Insert plan-usage percentages (Section 8).